Carles Trullàs

Data Scientist & ML Engineer

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carlestrullas carlestrullas

Profile

Data Scientist with a background in Mathematics, providing a very solid foundation in modeling, programming, and analytical thinking. Ranked first in class throughout high school, university, and graduate studies, showcasing strong motivation and achievement.

Education

MSc **Universitat Oberta de Catalunya**, Data Science — Sept 2023 – Jul 2026

- Specialized in Machine Learning, Deep Learning, and AI Agentic Frameworks to build autonomous systems.
- Honorable Mention in Machine Learning.

MSc **Polytechnic University of Barcelona**, Advanced Mathematics and Mathematical Engineering — Sept 2022 – June 2023

- The final qualification of the master's degree is 9.8 (best of the promotion).
- Awarded with 6 Honorable Mentions, including Master's Thesis.

BSc **Autonomous University of Barcelona**, Mathematics — Sept 2018 – June 2022

- *Extraordinary award* for the best academic record of the promotion.
- Awarded with 10 Honorable Mentions, including BSc final Thesis.

Experience

NTT Data. Business Advanced Analytics, Data Scientist & ML Engineer — Barcelona, Spain, Mar 2026 – Present

- Built a data-driven anomaly detection solution that identifies abnormal ordering patterns using ML-based demand forecasts, enabling early intervention before operational impact.

UVE Solutions, Data Scientist — Manresa, Spain, Sept 2024 – Mar 2026

- Designed and deployed an AI-powered search engine capable of finding Points of Sale with photos and reviews most similar to a given image or short text input.
- Delivered client-focused analytics to identify high-potential points of sale, supporting international companies in optimizing their commercial strategies.
- Collected data from multiple sources via scraping and performed ETL processes to produce clean, structured datasets ready for analysis and decision-making.

Autonomus University of Barcelona, Assistant Lecturer (Faculty of Mathematics) — Bellaterra, Spain, Sept 2023 – Aug 2024

- Taught courses in Statistics, Differential Equations, Linear Algebra, and Calculus.

Center of Mathematical Research (CRM), Scholarship — Bellaterra, Spain, Jan 2023 – Aug 2023

- I received a scholarship to work on my master's thesis in advanced mathematics.

Group on Dynamical Systems, UAB (GSDUAB), Research Intern — Bellaterra, Spain, July 2021 – Sept 2021

- Intern at the *Group on Dynamical Systems of UAB*, which belongs to the Mathematics Department at the Autonomous University of Barcelona.

Licenses & Certifications

Databricks Machine Learning Engineer Associate June 2026

Honors & Awards

Best Academic Record, Master's Degree in Advanced Mathematics July 2023

Extraordinary Award, Bachelor's Degree in Mathematics July 2022

Honorable Mention, Spanish Baccalaureate June 2018

Technologies

Programming: C, C++, R, Python, PySpark, Spark, SQL.

Microsoft Azure Ecosystem: Data Factory, Databricks, Azure Storage, Microsoft SQL Server & Azure Cosmos DB.

Vector DB: PostgreSQL.

Version Control & Repositories: Git, GitHub, Bitbucket.

Machine Learning & AI Platforms: Hugging Face.

IDEs: VS Code, Cursor, GitHub Copilot.

Data Engineering Tools: Docker, Kubernetes, Lens, Azure DevOps, Kibana (Elastic), RabbitMQ, FastAPI.

Project Management & Agile: Scrum, Agile Methodologies, Jira, Atlassian Ecosystem, Notion, Kanban.

Projects & Expertise Areas

ML-Based Anomaly Detection & MLOps Implementation: Developed a data-driven solution to proactively detect anomalous orders by combining machine learning, business rules, and scalable data pipelines. The system enables early identification of risks and prioritization of high-impact cases, improving operational efficiency and decision-making across the order-to-cash process.

- Developed an anomaly detection system leveraging supervised demand forecasting models to estimate expected order behavior at a customer–SKU–day level and identify statistically significant deviations.
- Built an end-to-end data pipeline integrating historical orders, contextual features, and feature engineering to support scalable outlier detection and classification of anomalous patterns.
- Established key MLOps practices such as model retraining cycles, champion–challenger evaluation, and CI/CD pipelines, enabling continuous improvement and robust performance in production.

Search AI Engine: Designed and deployed an *AI-powered search engine* capable of finding Points of Sale with photos and reviews most similar to a given image or short text input.

- Applied CLIP models (e.g., *SigLIP*, *Jina*) to generate embeddings for images and reviews.
- Utilized *GPUs* to accelerate embedding computation and *vector databases (PostgreSQL + HNSW indexing)* for efficient similarity search.
- Deployed a *production-ready API on Kubernetes* to integrate the AI search engine with the company's main platform, making it accessible to users.
- Developed an internal *Streamlit-based frontend* for the Data Science team, also used for client demos.

Web Scraping: Designed and maintained robust scrapers to extract information from multiple websites using Python libraries such as *BeautifulSoup* and *Selenium*. Ensured scalability, reliability, and adaptability to website changes.

Data Pipelines & ETL: Designed and implemented complex *ETL pipelines* to process large volumes of raw data into clean and structured datasets. Leveraged tools such as *Azure Storage*, *Azure Data Factory*, *Databricks*, and *Azure Cosmos DB* to orchestrate and optimize workflows.

Client Ad-hoc Projects (CPG sector): Developed tailored analytics solutions for international clients in the *Consumer Packaged Goods (CPG)* industry. Built ML models (*Linear/Logistic Regression*, *Naive Bayes*, *Decision Trees*, *XGBoost*, *Neu-*

ral Networks, *Clustering*, etc.) to identify Points of Sale (PoS) with the highest potential or specific characteristics, supporting commercial strategy.

Publications

Quadratic differential systems with a WF1 and a finite saddle-node August 2023

International Journal of Bifurcation and Chaos. Joan C. Artés & ***Carles Trullàs***

Languages

Catalan and Spanish: Native proficiency.

English: Full professional proficiency.